

when is it your turn?

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 why it collapses
- 2 the clocked way
- 3 the clock-free way

part 1

why it collapses

how do eighty
people
stay together?

image: ai-generated (gpt-image-2)



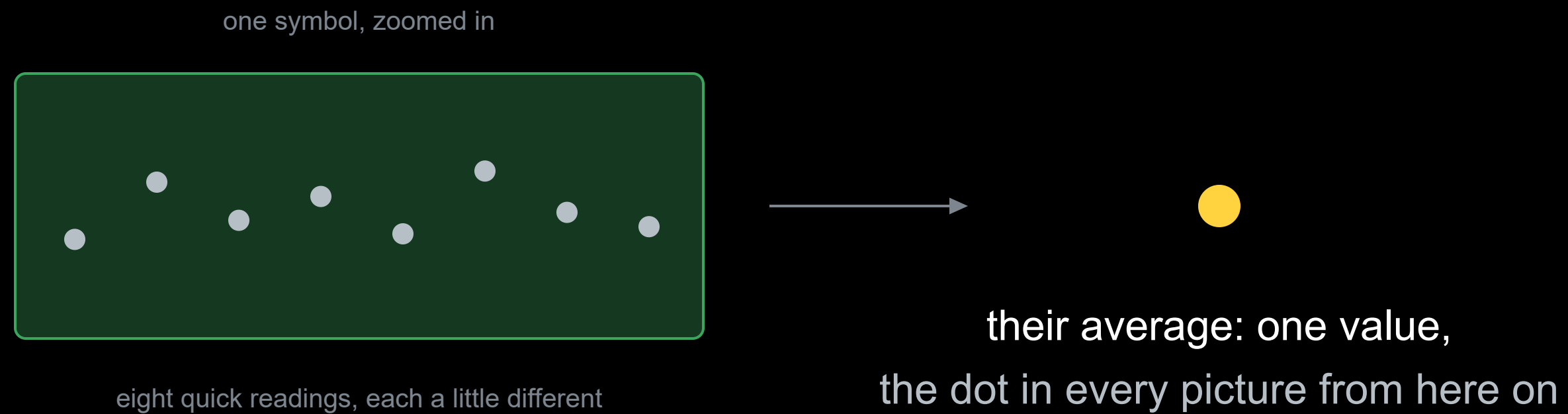
why it collapses

your homework: where did it tip?

team	symbols per second	what broke
team 7	14	letters shifted by one, slowly, after about a minute
team 2	9	suddenly nothing matched a profile any more
team 11	6	every second word was garbled from the start
team 5	4	held, we did not dare go faster

same limit for everyone. very different numbers.

what one dot means on the next slides



one dot in the pictures = many readings, averaged.

cause one: no two clocks run alike

sender: one colour per slot

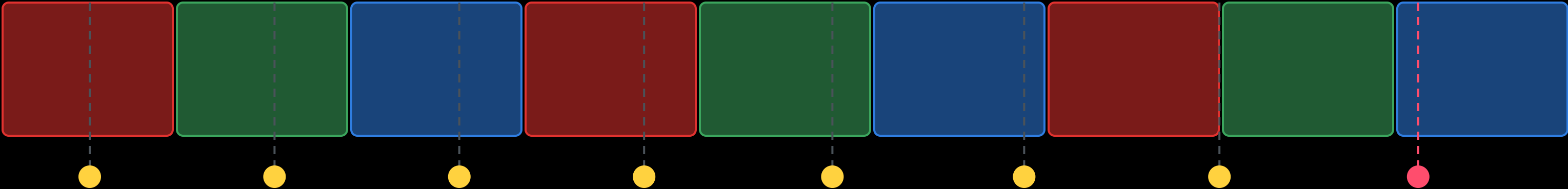


receiver: reads at

the tiny error per symbol is not the problem. the sum is.

cause one: no two clocks run alike

sender: one colour per slot

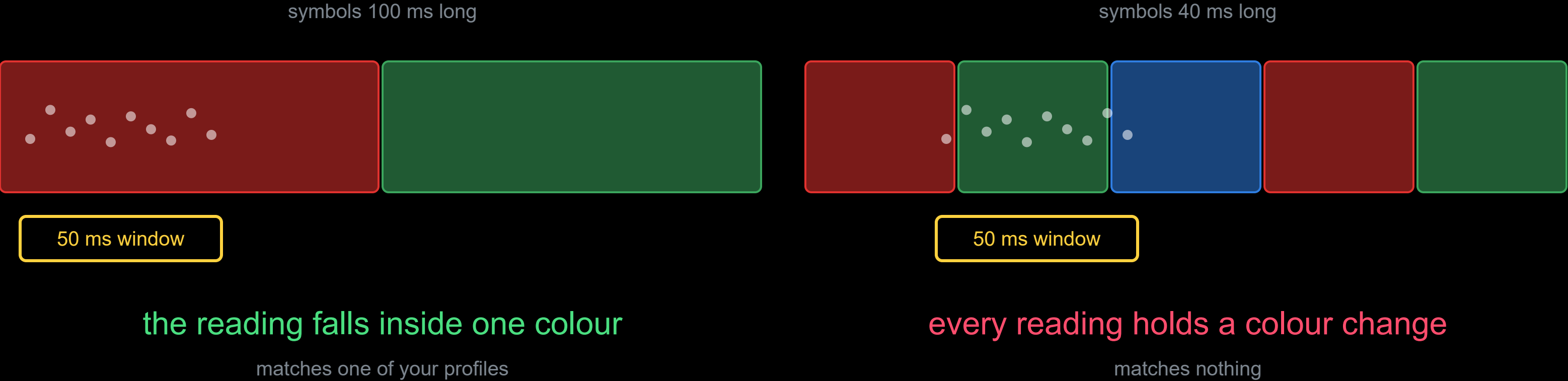


receiver: reads at

in the wrong slot

the tiny error per symbol is not the problem. the sum is.

cause two: the sensor needs time to look



no error, no warning, just a meaningless number.

both causes are one question

when does a new symbol begin?

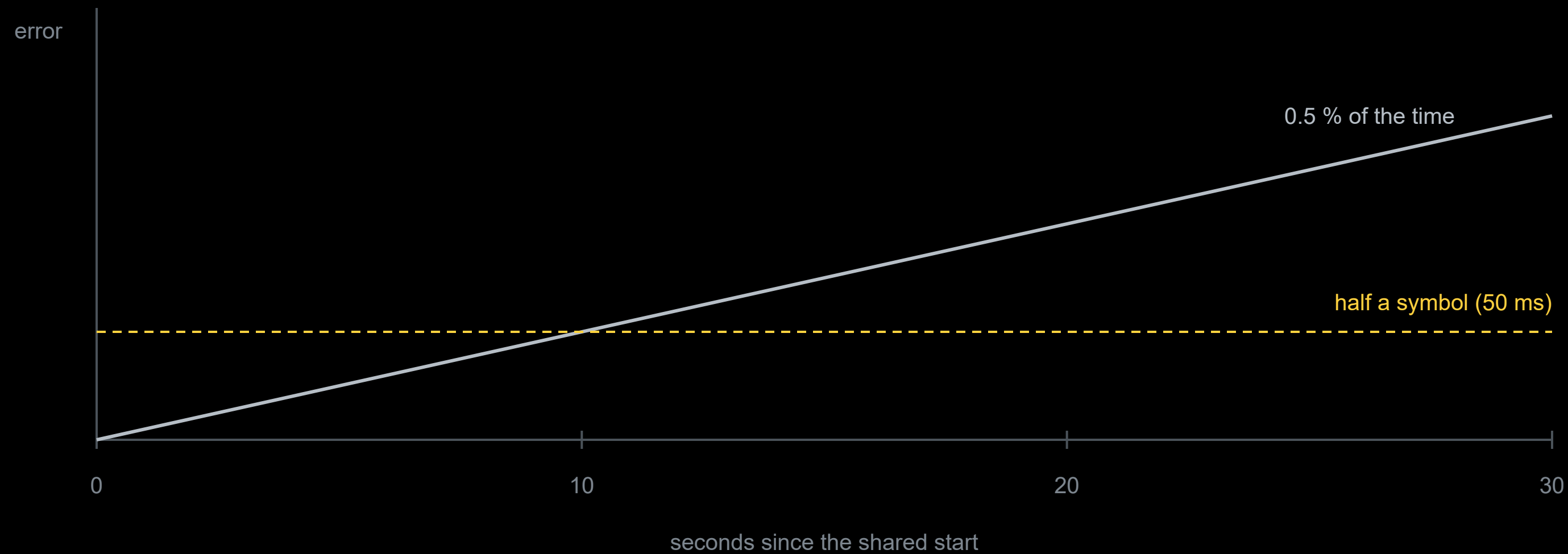
answer one: time tells you.

answer two: change tells you.

part 2

the clocked way

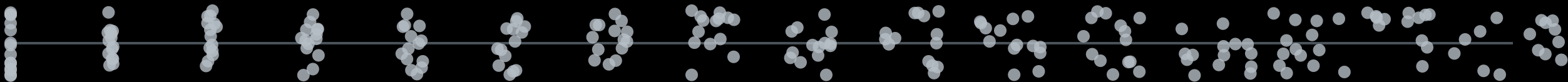
when does it tip?



a shared start is not a question of if it tips, only when.

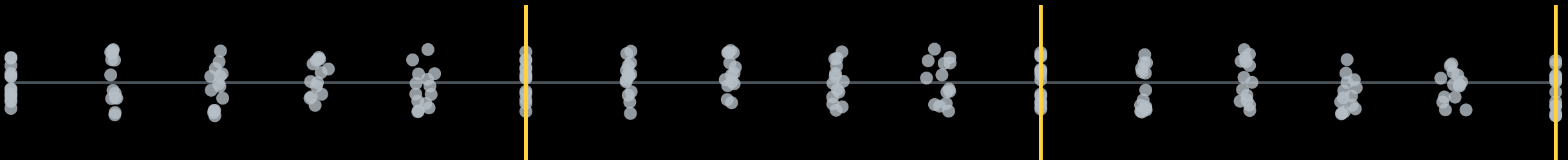
eighty people, one clap

one start



one clap = one cloud of people. the cloud gets wider.

a pulse every 5 s



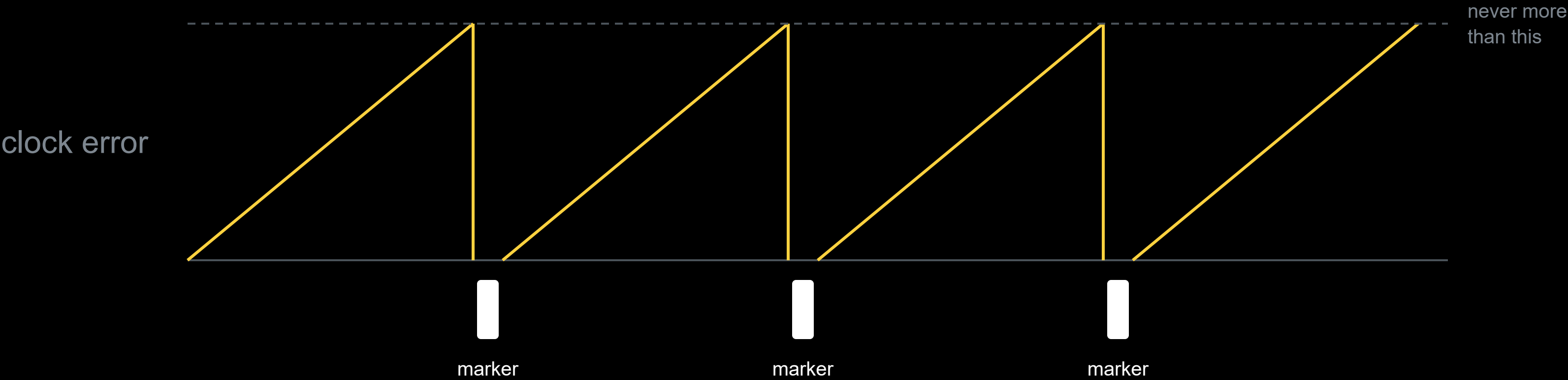
every pulse pulls everyone back together

0 5 10 15

seconds

agreeing once is not enough. agreeing again and again is.

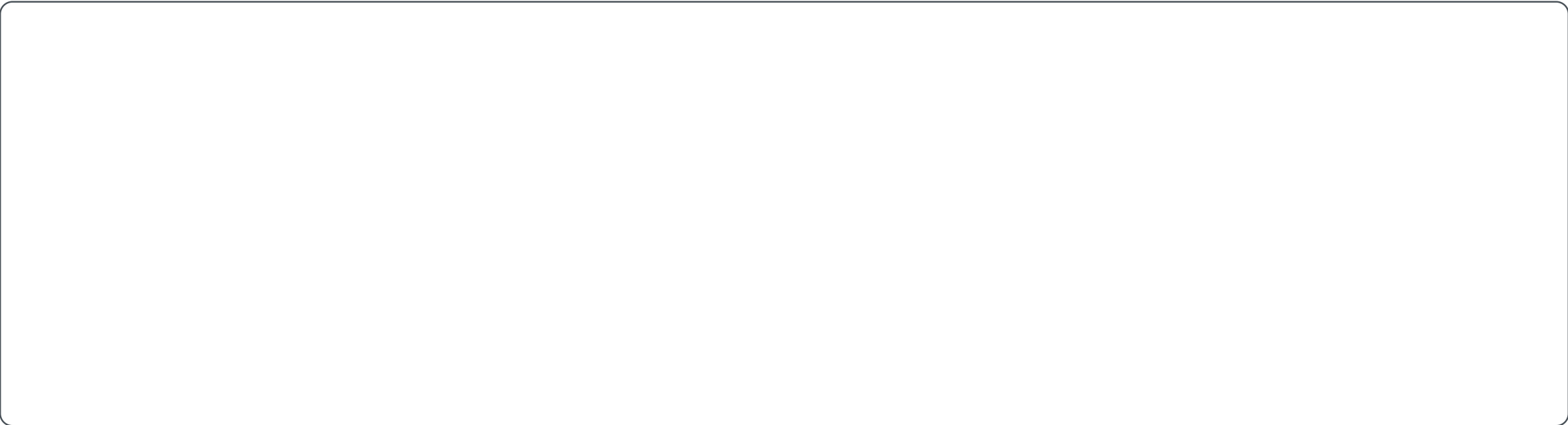
recurring markers reset the drift



a marker is a symbol reserved for this job: a colour outside your data alphabet, or a short dark pause

the error can never grow beyond one block.

try it: the drift simulator



:sent
:received

simulated clocks: real crystals differ by fractions of a percent. the effect is the same, only slower.

receiver clock
+2 %
runs slow: samples too late

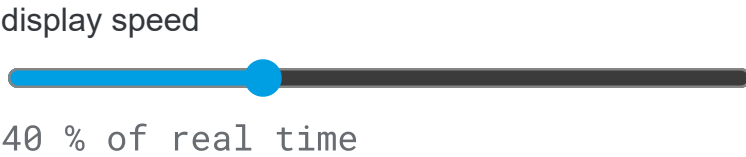
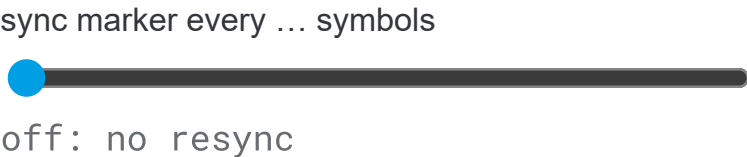
currently off by
+0.00
slots
half a slot means wrong reads

half a slot off after
25 sym
then reads start to miss

wrong reads

of the last 100 symbols

marker overhead
0 %
slots spent on markers



pause

sample once

send marker

reset

the price of the clocked way

start once

not a single symbol spent

drift adds up without limit

recurring markers

drift stays bounded

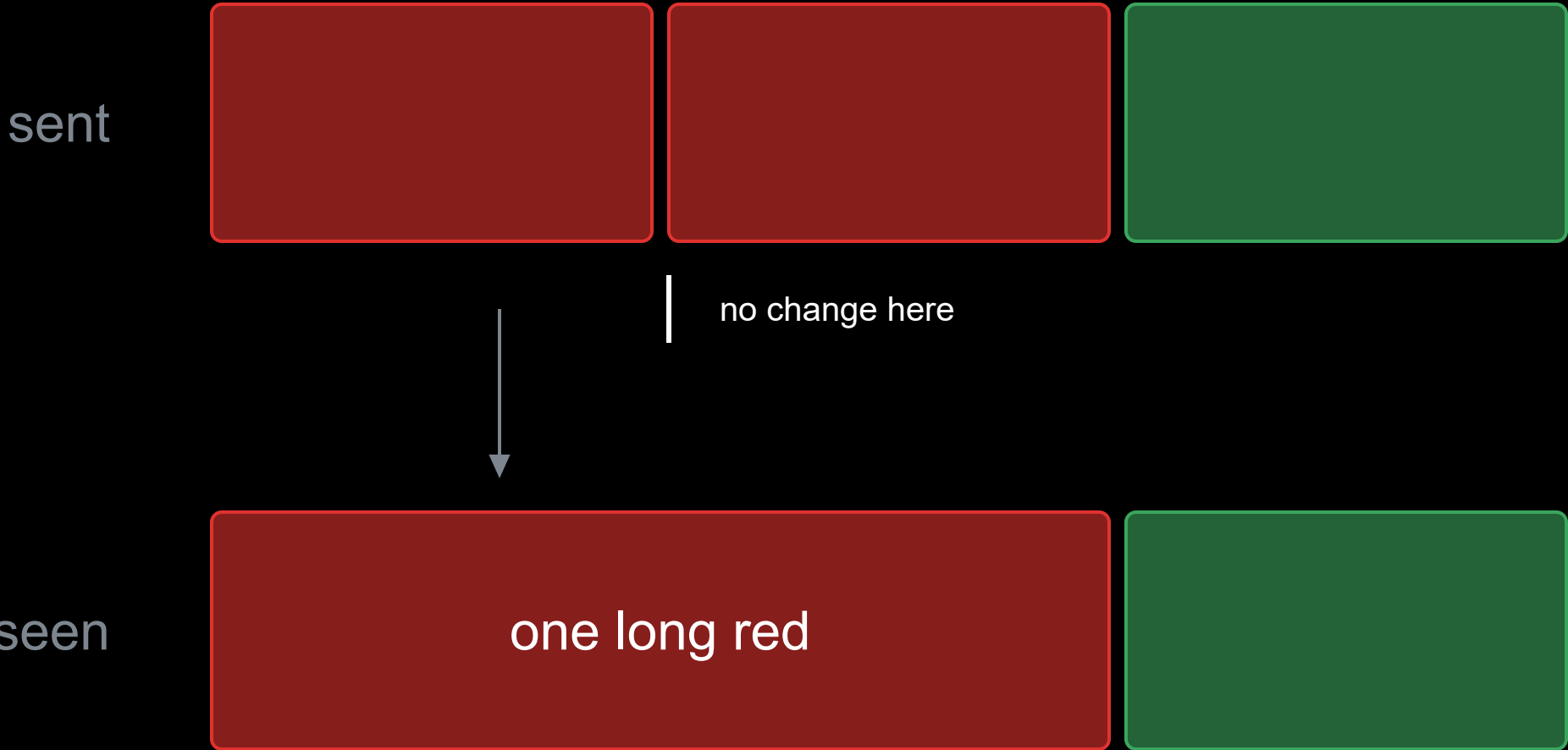
markers carry no message

how bad are your clocks? that is a measurement, not a belief.

part 3

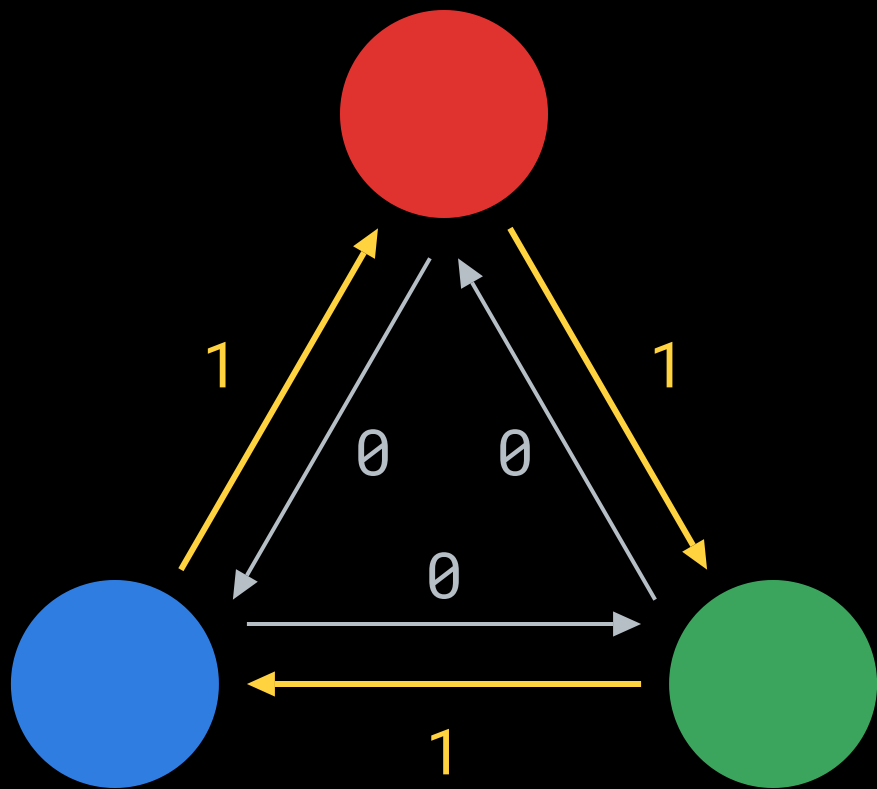
the clock-free way

first idea: a change means "next symbol"



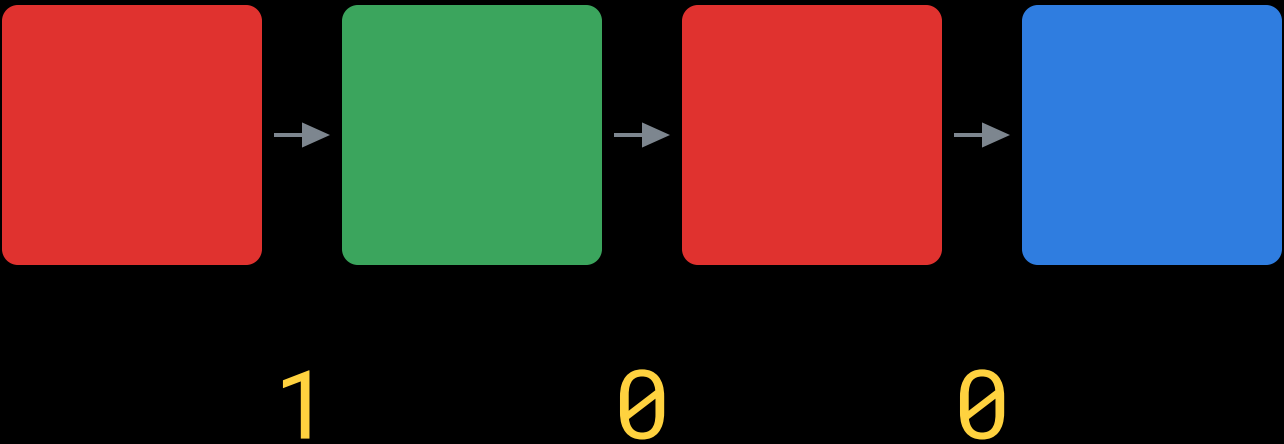
no change, no symbol. "red red" arrives as one.

the fix: the transition carries the bit



clockwise: 1, anticlockwise: 0

your own colour is never allowed next



measure as often as you like. decode only on change.

freedom is paid in capacity

bits carried by one symbol

k	as states	as transitions
3	1.58	1.00
4	2.00	1.58
5	2.32	2.00

five colours,
four allowed successors,
a clean 2 bits

want a clean 2 bits per transition? you need five colours, not four.

the most common wrong conclusion

clock-free is not time-free

how long must a colour hold to count?

how many transitions per second can the sensor follow?

what your file costs in minutes

$$t = n / (r * b)$$

n bits to send

r symbols per second

b bits per symbol

two levers, nothing else: bits per symbol, symbols per second.

what your file costs in minutes

$$t = n / (r * b)$$

n bits to send

r symbols per second

b bits per symbol

16 000 bits, 4 colours, 5 symbols per second

clocked $b = \log_2(k) = 2.00$

about 27 minutes

clock-free $b = \log_2(k-1) = 1.58$

about 34 minutes

two levers, nothing else: bits per symbol, symbols per second.

today

- 1 the clapping experiment
- 2 workshop: find your maximum rate, with evidence
- 3 write down which way you lean, clocked or clock-free, and why

no two clocks
agree.
the only choice is
what you do
about it.

